

1. You are given the following matrix A and its reduced row echelon form R .

$$A = \begin{bmatrix} -1 & 0 & -1 & 2 \\ -3 & 1 & -1 & 13 \\ -1 & 2 & 0 & 10 \\ -1 & 1 & -1 & 5 \\ 3 & 0 & 2 & -8 \end{bmatrix} \quad R = \begin{bmatrix} 1 & 0 & 0 & -4 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

(a) (4 points) Write the general solution to $A\mathbf{x} = \mathbf{0}$ in parametric vector form.

(b) (2 points) Find the specific solution to $A\mathbf{x} = \mathbf{0}$ for which $x_1 + 6 = x_2$.

(c) (2 points) It is given that $\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$ is a solution to $A\mathbf{x} = \mathbf{c}$. Find $\mathbf{c}$.

(d) (1 point) Find the dimension of $\text{null}(A)$.

(e) (2 points) What is the rank of A ?

(f) (2 points) Let $\mathbf{a}_1$, $\mathbf{a}_2$, $\mathbf{a}_3$, and $\mathbf{a}_4$ denote the columns of A . Write the first column of A as a linear combination of the other three columns of A .

2. Let $A = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 1 & 2 \\ 3 & 2 & 3 \end{bmatrix}$

(a) (4 points) Find the inverse of A .

(b) (3 points) Use your answer in part (a) to find the matrix B , if $BA + A^T = \begin{bmatrix} 0 & 2 & 1 \\ 2 & -7 & -1 \\ 7 & -5 & 3 \end{bmatrix}$.

3. (4 points) Find the only value of a such that the following matrix equation has infinitely many solutions.

$$\begin{bmatrix} 1 & a-4 \\ a & a-6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -a-1 \\ -12 \end{bmatrix}$$

4. Let A , B , and C be 3×3 matrices such that $\det(ABC) = 0$, $\det(2B^T C) = 6$, $\det(C^{-1}) = 2$.

(a) (2 points) Find $\det(A)$.

(b) (2 points) Find $\det(B)$.

(c) (2 points) Find $\det(C)$.

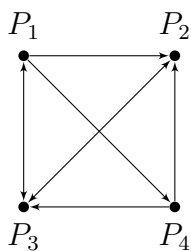
5. (5 points) The determinant of the following matrix A is 2026. Find k .

$$A = \begin{bmatrix} 1 & 0 & 3 & 1 \\ 2 & 0 & 7 & 2 \\ 3 & k & 8 & 11 \\ 3 & 0 & 2 & 5 \end{bmatrix}$$

6. (4 points) You are given $A = \begin{bmatrix} 7 & -5 & 0 & 4 \\ -3 & 3 & -6 & 9 \\ 5 & 1 & 1 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -1 & 2 & -3 \\ 7 & -5 & 0 & 4 \\ 5 & 1 & 1 & -1 \end{bmatrix}$

Find two elementary matrices E_1 and E_2 such that $B = E_2 E_1 A$. (There is more than one correct answer, but you only need to find one pair of matrices that work.)

7. Let $A = \begin{bmatrix} 3 & 1 & -1 \\ -4 & -2 & 1 \\ 2 & 0 & -1 \end{bmatrix}$.
- (a) (4 points) Find the characteristic polynomial of A .
- (b) (3 points) Find all eigenvalues of A . (You do not need to find eigenvectors.)
8. (6 points) It is given that $A = \begin{bmatrix} -1 & -4 & 4 \\ 0 & 1 & 0 \\ -2 & -4 & 5 \end{bmatrix}$ has two eigenvalues: $\lambda_1 = 1$ and $\lambda_2 = 3$. Find a diagonal matrix D and an invertible matrix P such that $AP = PD$.
9. (4 points) Let $A = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 2 & 1 & 4 & -4 \\ 1 & 4 & 2 & 2 \\ 0 & 3 & 0 & 2 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$. It is given that $\det(A) = 10$. Using Cramer's Rule, find x_4 only in the solution to $A\mathbf{x} = \mathbf{b}$.
10. (3 points) Let $H = \{\mathbf{x} \in \mathbb{R}^3 : x_1 \leq x_2 \leq x_3\}$. Show that H is **not** a subspace of $\mathbb{R}^3$.
11. (5 points) Let $H = \left\{ \mathbf{x} \in \mathbb{R}^4 : \mathbf{x} \text{ is orthogonal to } \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} \right\}$. Find a basis for H .
12. (6 points) Provide specific examples of the following:
- (a) Vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^3$ such that $\text{span}\{\mathbf{u}, \mathbf{v}\}$ is not a plane.
- (b) A 2×2 matrix A such that $A \neq 0$ but $A^2 = 0$.
- (c) Vectors $\mathbf{x}, \mathbf{y}, \mathbf{z}$ in $\mathbb{R}^3$ such that $\{\mathbf{x}, \mathbf{y}\}$ and $\{\mathbf{y}, \mathbf{z}\}$ are linearly dependent sets, but $\{\mathbf{x}, \mathbf{z}\}$ is a linearly independent set.
13. (6 points) Let $\mathcal{L}$ be the line $\mathbf{x} = \begin{bmatrix} 5 \\ 4 \\ 1 \end{bmatrix} + t \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$ for $t \in \mathbb{R}$. Let P be the point $(-2, 7, 0)$. Find the point on the line $\mathcal{L}$ that is nearest to point P .
14. You are given the following four points: $P(-3, -3, 8)$, $Q(0, -5, 6)$, $R(1, -4, 2)$, and $S(4, -1, 5)$.
- (a) (2 points) Find a parametric vector equation of the line containing P and Q .
- (b) (4 points) Find an equation of the plane containing P and Q that is parallel to the line containing R and S . Your answer must be in the form $ax + by + cz = d$.
- (c) (2 points) Find the area of the triangle whose vertices are P , Q , and R .
- (d) (2 points) Find the cosine of the angle formed by vector $\overrightarrow{PQ}$ and vector $\overrightarrow{PS}$.
15. You are given the following directed graph.



(a) (4 points) Write the adjacency matrix M for the graph.

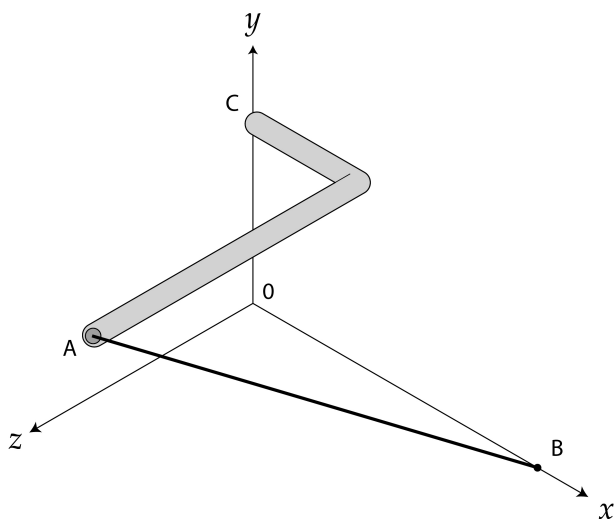
(b) The third row of M^4 is provided below.

$$M^4 = \begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ 2 & 4 & 5 & 2 \\ \cdot & \cdot & \cdot & \cdot \end{bmatrix}$$

i. (2 points) How many **walks of length 4** start at P_3 and end at P_4 ?

ii. (2 points) How many **closed walks of length 5** start at P_3 ?

16. (6 points) The following diagram shows a bent metal bar and a cable (tied between A and B). The tension in the cable is 18N. The points shown are $A(3, 4, 7)$, $B(7, 0, 0)$, and $C(0, 4, 0)$, measured in meters. Find the torque exerted at A about C .



Answers

1. (a) $\mathbf{x} = t \langle 4, -3, -2, 1 \rangle$, where $t \in \mathbb{R}$, (b) $\mathbf{x} = -\frac{6}{7} \langle 4, -3, -2, 1 \rangle$, (c) $\mathbf{c} = \langle 0, 10, 11, 4, -3 \rangle$, (d) 1, (e) 3, (f) $\mathbf{a}_1 = \frac{3}{4}\mathbf{a}_2 + \frac{1}{2}\mathbf{a}_3 - \frac{1}{4}\mathbf{a}_4$
2. (a) $A^{-1} = \begin{bmatrix} -1 & -5 & 4 \\ 0 & -3 & 2 \\ 1 & 7 & -5 \end{bmatrix}$, (b) $B = \begin{bmatrix} -1 & -9 & 6 \\ -2 & 8 & -5 \\ -5 & -4 & 6 \end{bmatrix}$
3. 3
4. (a) 0, (b) $\frac{3}{2}$, (c) $\frac{1}{2}$
5. -1013
6. $E_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $E_2 = \begin{bmatrix} -\frac{1}{3} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
7. (a) $-\lambda^3 + \lambda$, (b) $-1, 0, 1$
8. $D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix}$, $P = \begin{bmatrix} -2 & 2 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$
9. $\frac{3}{5}$
10. We show that closure under scalar multiplication fails. Let $\mathbf{u} = \langle 1, 2, 3 \rangle$, which is in H . Let $k = -1$. We see that $k\mathbf{u} = \langle -1, -2, -3 \rangle$, which is not in H , since $-1 > -2 > -3$.
11. $\mathcal{B} = \{ \langle -2, 1, 0, 0 \rangle, \langle -3, 0, 1, 0 \rangle, \langle -4, 0, 0, 1 \rangle \}$
12. (a) $\mathbf{u} = \langle 1, 1, 1 \rangle, \mathbf{v} = \langle 2, 2, 2 \rangle$
 (b) $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$
 (c) $\mathbf{x} = \langle 1, 0, 0 \rangle, \mathbf{y} = \langle 0, 0, 0 \rangle, \mathbf{z} = \langle 0, 1, 0 \rangle$
13. $(\frac{14}{3}, \frac{10}{3}, \frac{1}{3})$
14. (a) $\mathbf{x} = \langle -3, -3, 8 \rangle + t \langle 3, -2, -2 \rangle$, where $t \in \mathbb{R}$, (b) $-y + z = 11$, (c) $\frac{15}{2}$, (d) $\frac{23}{\sqrt{17}\sqrt{62}}$
15. (a) $M = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix}$, (b.i) 2, (b.ii) 8
16. $\langle 56, 98, -24 \rangle$ Nm